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TCN-TOPSIS model and its application in digital twin system for rocket safety status prediction

Fugang Li, Bo Li, Lei Zhao, Hao Li, Yuehui He, Junsheng Zheng, Yonglin Bao and Zhiwei Ding

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ABSTRACT

Rocket flight experiments are crucial for analysing and validating the performance of the carried mechanical equipment. However, due to the difficulty of processing a large amount of real-time monitoring data, achieving a comprehensive safety assessment of the flying rocket remains challenging. A digital twin system based on the proposed TCN-TOPSIS was designed to conduct real-time twin safety assessment of the flying rocket, while providing intelligent decision assistance. TCN network was used to predict the rocket flight parameters, and TOPSIS model is used to conduct real-time safety assessment. The digital twin system, designed with the proposed TCN-TOPSIS model in this paper, accurately predicted the rocket safety status in less than 3 ms, outperforming RNN, GRU, and LSTM. The result indicated that, the proposed TCN-TOPSIS model is a potentially effective and accurate method of digital twin system for rocket safety status prediction.

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1. Introduction

As a carrier platform, the rocket is often used in preliminary experiments to test whether the carried electrical system and mechanical structure work well. The rocket carries tons of fuel at an extremely high speed, and unexpected conditions may sometimes occur during the flight process. If rocket flight parameters (Flight position, acceleration, velocity, and flight attitude) exceed predicted thresholds, the rocket's drop location may become uncontrollable, which can cause unexpected damage if not handled promptly and appropriately. With extremely high flight speeds, it is necessary to monitor flight parameters in real-time and terminate the rocket's power system and destroy it promptly if an unexpected condition occurs.

The rocket safety decision-making system is a computer-based system that integrates ballistic data processing and safety control decision-making. The system compares flight parameters with pre-stored theoretical ballistic parameters. If the deviation value falls within the pre-set fault range, the rocket is considered to fly safely. When actual flight parameters reach (or exceed) the fault range, experts and technicians need to synthesise various parameters for further reasoning and judgement, a process that usually takes a long period of time. If analysis results reach the allowable explosion threshold, a termination command must be issued and executed to prevent the rocket from flying out of the allowable range and causing further damage. Due to the extremely fast flight speed, the whole analysis and decision process must be highly real-time to ensure termination command can be made and implemented promptly. With many flight parameters and a complex reasoning mechanism, experts and technicians must issue instructions within a very short time. However, in reality, it often takes a long time from receiving monitoring data to sending instructions to the rocket.

Initially, the safety decision-making process depended mainly on experienced experts and operational guidelines. Sandia National Lab (SNL) [1] and the U.S. Space and Missile Systems Center (SMSC) [2]

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presented a general set of guidelines for addressing safety issues related to rocket flight states. Hope [3] described a set of sequential steps to identify, evaluate, and compare hazards and their associated risks. Systems-Theoretic Process Analysis (STPA) was used to evaluate launch vehicle safety [4–6]. Chai Yi [7] and Lawrence [8,9] developed a description model of launch vehicle flying status. However, due to the complexity of rocket structure and the coupling of flight data, the accuracy and timeliness of safety control instructions remain unsatisfactory, and meeting actual needs remains challenging. To enhance the efficiency and accuracy of safety decision-making, computer expert systems were developed to replace or augment human thinking and decision-making processes [10,11]. The computer system integrates expert experience with book knowledge [12], uses computational power to simulate the thinking and judgement processes of experts [13,14], and can quickly assess flight statuses of rockets based on monitored data [15,16]. This computer analysis system can directly analyse various parameters [17], reducing decision-making time and greatly minimising the probability of damage [18]. Concurrently, prognostic and health management (PHM) methodology has been developed [19].

Since the 1990s, data-driven prediction theory has made rapid progress. The goal is to know fault information in advance before rocket trajectory anomaly appears. Rocket trajectory prediction (TP) is a challenging, inherently data-driven time-series modelling problem. Rocket flight status is predicted using current flight data, providing an early warning function for security control decision-making. Main methods used include real-time trend prediction and fuzzy mathematics, which enable real-time prediction and good operability, such as multi-stage hybrid approach [20], modelling methodology [21], continuation and bifurcation methods [22], ARMA Model [23]. In recent years, data-driven neural network methodologies [19,24,25] are increasingly used for rocket trajectory prediction, such as Convolutional Recurrent Neural Networks [26,27], Deep spatio-temporal neural networks [28], hybrid-recurrent networks [29], Social Long Short-Term Memory (S-LSTM) network [30], Deep Long Short-Term Memory Networks [31], hybrid CNN-LSTM model [32,33], Rnn-cnn hybrid model [34], CNN-GRU model [35] and Att-BiLSTM [36]. Rocket flight status prediction is inherently a data-driven time-series modelling problem. Recurrent and Convolutional architectures have a long history in sequence prediction. The current ‘default’ mindset in much of the deep learning community is that generic sequence modelling is best handled using recurrent networks [37]. Recurrent networks broadly refer to networks that maintain a vector of hidden activations, which are propagated through time. Because recurrent networks propagate a hidden state forward, they are theoretically capable of infinite memory, and can make predictions based on data that occurred arbitrarily long ago. This possibility is realised even more so for the now-standard architectures of Long Short-Term Memory networks (LSTMs) [38], or recent incarnations such as the Gated Recurrent Unit (GRU) [39]. Jozefowicz et al. [40] concluded that if there were ‘much better architectures than the LSTM’, then they would be ‘not trivial to find’. However, recent years have seen a re-emergence of convolutional models for sequence data. Perhaps most notably, the WaveNet [41] applied a stacked convolutional architecture and was additionally applied to sequence modelling. LeCun et al. [42] firstly proposed and examined the usage of CNNs on time-series data, pointing out that the same kind of feature extraction could work well on sequence modelling with convolutional filters. Research [43] shows that on a variety of sequence modelling tasks, including many frequently used as benchmarks for evaluating recurrent networks, the temporal convolution network (TCN) outperforms baseline RNN methods (LSTMs, GRUs, and vanilla RNNs) and sometimes even highly specialised approaches. For example, TCN has been applied to mechanical axis health assessment and status prediction [44] and industrial system health prediction [45,46].

Recently, the development of semi-physical simulation and digital twin has improved the theory of equipment testing and verification through digitalisation and physical experiments [47–49]. Grieves [50] initially proposed the digital twin model, while Tao Fei proposed the five-dimensional model of digital twin [49] and the architecture of digital experimental testing and verification system [51,52]. As an effective means of achieving information-physical fusion, digital twin simultaneously utilises 3D models, data, and actual equipment to improve the efficiency of equipment performance verification experiments and enhance data-driven decision support capabilities [53]. Compared to traditional digital technologies, digital twin emphasises deep integration of digital and physical equipment, achieving real-time

interaction and data consistency, thereby, providing more accurate analysis and decision-making services [54].

Due to the parallel computing capabilities and high prediction accuracy, TCN was adopted in this article to predict rocket flight parameters. Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) model was employed for real-time assessment of rocket safety status. A digital twin system has been developed to facilitate this real-time TCN-TOPSIS decision assistance. In essence, the TCN-TOPSIS model uniquely combines the high-speed, high-accuracy temporal forecasting of TCNs with the systematic, multi-parameter decision-making logic of TOPSIS. This synergy directly tackles the core challenges of timeliness and reasoning complexity in rocket safety, offering a more integrated and computationally efficient solution than a purely predictive PHM approach or a potentially slower, less interpretable CNN-LSTM model. Furthermore, recent advances in time-series forecasting for safety-critical systems, such as real-time landslide detection [55], have reinforced the importance of developing models that balance high accuracy with computational efficiency for immediate decision-making – a core objective of the TCN-TOPSIS framework proposed herein.

2. Related work

Rocket flight status prediction is essentially a data-driven time series modelling problem. It requires timely and accurate prediction results, demanding a simple prediction model with fast calculation speed.

2.1. Sequence modelling

Before discussing rocket flight status prediction, sequence modelling needs to be introduced. Given an input sequence $x_0, \dots, x_T$, the task to predict some corresponding outputs $y_0, \dots, y_T$ at each time step. Crucially, when predicting the output y_t for time t , only previously observed inputs $x_0, \dots, x_t$ can be used. Formally, sequence modelling network is a function $f: X^{T+1} \rightarrow Y^{T+1}$ that induces the mapping:

$$\hat{y}_0, \dots, \hat{y}_T = f(x_0, \dots, x_T) \quad (1)$$

If y_t depends only on $x_0, \dots, x_t$ and not on any future inputs $x_{t+1}, \dots, x_T$, satisfying the causal constraint. The goal of sequence modeling is to find a network f that minimises some expected loss between actual outputs and predictions $L(y_0, \dots, y_T, f(x_0, \dots, x_T))$. Sequences and outputs are drawn according to some distribution. This formalism encompasses many settings, including auto-regressive prediction where the target output is simply the input shifted by one time step.

2.2. Recurrent neural networks

In deep learning, sequence modelling is closely associated with recurrent neural networks (RNN). The title of the sequence modelling chapter in the authoritative deep learning textbook is ‘Sequence Modelling: Recurrent and Recursive Nets’ [56], underscored the belief that sequence modelling is intrinsically linked to recurrent architectures.

Recurrent networks are specialised sequence models that maintain a hidden activation vector propagated through time [57–59]. This architecture family has gained immense popularity due to prominent applications in language modeling [60–62] and machine translation [63,64]. Basic RNN architectures are notoriously difficult to train [65,66] and more elaborate architectures such as LSTM [38] and GRU [67] are commonly used instead. Numerous advanced regularisation and optimisation schemes for LSTMs have been proposed [68–70], significantly advancing accuracy on some datasets. Many other architectural innovations and training techniques for recurrent networks continue to be explored [71–79].

Many empirical studies have evaluated the effectiveness of different recurrent architectures, motivated in part by the numerous design freedoms in these architectures. Chung et al. [80] compared different recurrent unit types (LSTM vs GRU) on polyphonic music modelling. Pascanu [81] explored

different deep RNN constructions and evaluated architectures on polyphonic music modelling and character/word-level language modelling. Zhang et al. [82] and Melis [83] systematically analysed RNN connection architectures and evaluated on character-level language modelling and synthetic stress tests, concluding that ‘LSTMs outperform more recent models’. Jozefowicz et al. [40] searched over ten thousand different RNN architectures and evaluated performance on various tasks, concluding that if ‘significantly better architectures’ existed, they would be ‘hard to find’. Greff et al. [84] benchmarked eight LSTM variants on speech recognition and polyphonic music modelling, also finding no ‘significant improvement over the standard LSTM architecture’.

Multiple comprehensive evaluations of RNN architectures on representative sequence modelling tasks exist. Convolutional networks [85] have been applied to sequences for decades [86]. Yin et al. [87] reported a comparison of convolutional recurrent networks for sequence modelling tasks. Sequence modelling requires architectures that can process whole sequences. Convolutional architectures, recently successful, were once weaker due to the absence of elements like dilated convolutions and residual connections.

2.3. Temporal convolution network

As an important method for solving the rocket flight status prediction problem, the computational cost of the RNN model grows exponentially with time. Shaojie Bai [43] compared generic convolutional and recurrent architectures on sequence modelling tasks, using both synthetic stress tests and real-world datasets from multiple domains. Results showed a simple convolutional architecture outperforms recurrent networks like LSTMs, across diverse tasks and datasets, while having longer effective memory.

Subsequently, a temporal convolution network (TCN) architecture is proposed for representing convolutional networks. This architecture is informed by recent research, but is deliberately kept simple, combining some of the best practices of modern convolutional architectures. Systematic comparisons with canonical recurrent architectures like LSTMs and GRUs on sequence modelling tasks reveal that TCNs convincingly outperform baseline recurrent architectures across a broad range of sequence modelling tasks. The TCN architecture appears more accurate than traditional recurrent networks like LSTMs and GRUs, and is also simpler and clearer. It has therefore been a more appropriate starting point in applying deep networks to sequential data. Therefore, TCN is adopted to predict rocket flight parameters in digital twin system.

2.4. Safety assessment

Rocket safety status assessment was carried out using an evaluation model, based on predicted flight parameters. TOPSIS, introduced by Hwang and Yoon [88], is a powerful and flexible comprehensive evaluation method. The basic idea of the TOPSIS model is straightforward [89]. As a multi-parameter assessment model, this methodology reduces multiple parameters into a single overall value (safety score) to facilitate rocket safety assessment of different flight conditions [90]. It originates from the concept that the higher safety score should have the shortest distance from the positive-ideal working condition A^+ and the farthest from the negative-ideal working condition A^- [91].

2.5. Method review

Recent advances in time series forecasting across various domains provide valuable methodological insights for rocket trajectory prediction. The hybrid approach for air quality forecasting [92] demonstrates the effectiveness of combining statistical analysis with deep learning architectures. Similarly, the timezone-aware auto-regressive LSTM model [93] highlights the importance of capturing periodic patterns and temporal dependencies in sequential data. These works collectively emphasise the value of thorough temporal analysis and hybrid modelling approaches, which informed the methodological decisions in this study.

In summary, due to the extremely fast flight speed, timely prediction of rocket status is necessary. The common RNN model cannot meet real-time prediction requirements. As TCN models substantially

outperform generic recurrent architectures such as LSTMs and GRUs, it can be adopted to predict rocket flight parameters. Subsequently, real-time assessment of rocket safety status can be conducted by TOPSIS model in digital twin system.

3. Methods

3.1. TCN-TOPSIS model

The architecture of the proposed TCN-TOPSIS model, as illustrated in Figure 1, comprises two parts. In the first part, rocket flight parameters (or characteristics) were predicted through a TCN model. Subsequently, the safety status of the rocket is evaluated by the TOPSIS model. It is worth noting that the TCN-TOPSIS model is processed in real-time during rocket flight in digital twin system.

3.2. Rocket flight parameters prediction

The main ingredient of TCN (Figure 2) is causal convolution which is able to look back at historical data [94]. The joint probability of flight parameters $x = \{x_1, \dots, x_T\}$ is factorised as conditional probabilities $p(x|h) = \prod_{t=1}^T p(x_t|x_1, \dots, x_{t-1}, h)$. Given rocket condition h , each output x_t is conditional at all previous outputs. By using causal convolution with 1D convolutional kernels, conditional predictions can be made with chronological order and in parallel. After each sample is predicted, it is fed back into the network to predict the next sample. Since causal convolutions do not have recurrent connections, they are typically faster to train than RNNs. Alternatively, causal convolution requires extremely deep layers to increase the receptive field when applied to long sequences like rocket flight data. To solve this problem, dilated convolution is applied, which enables the network to have exponentially large receptive fields with just a few layers. The effective receptive field of one layer is $(k-1)d$, with filter k

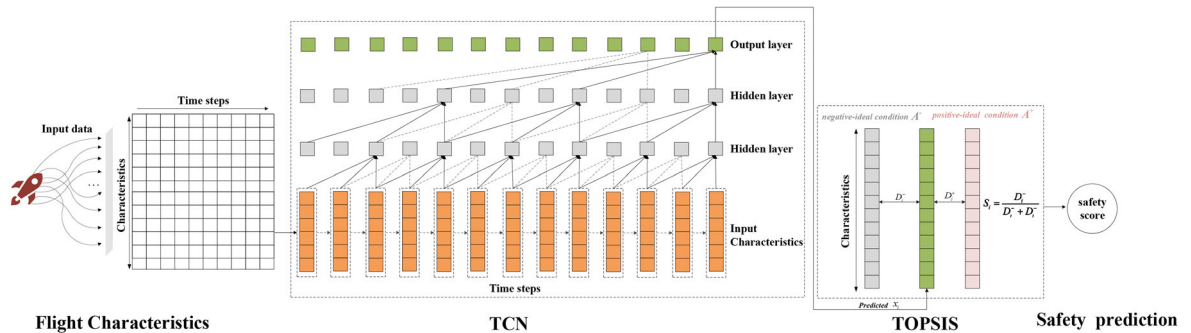


Figure 1. Architecture of the proposed TCN-TOPSIS model, which includes two main parts: (A) prediction of rocket flight parameters based on TCN; (B) Rocket safety status assessment based on TOPSIS.

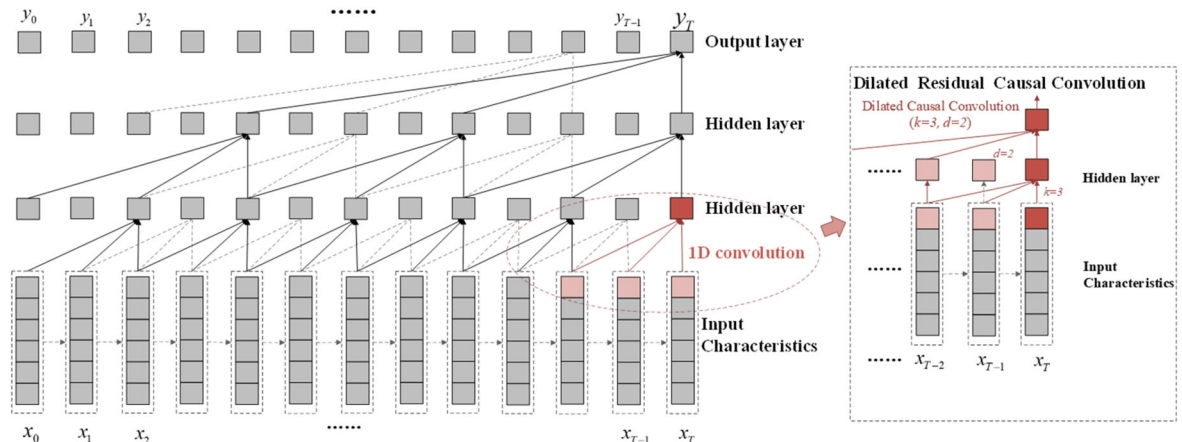


Figure 2. Architecture of the TCN model proposed in this paper.

and dilation d . The dilated convolution allows the network to operate on a coarser scale by increasing d exponentially with depth, $d = O(2^i)$ at level i . Importantly, instead of the entire transformation $F(x_1, \dots, x_t)$, residual connections are employed to allow layers to learn modification to the identity mapping, which can be described as $x_{out} = \text{Activation}(x_t + F(x_1, \dots, x_t))$.

Since the same filter is used in each layer of TCN, convolution computation can be parallelised, enabling processing of long input sequences (flight parameters) in their entirety. Semi-physical simulations of rocket flight have generated large amounts of time series data. Our work aims to enable timely predictions during future rocket flight experiments. Appropriate TCN architectures were identified by tuning hyperparameters, and a suitable performance evaluation metric was selected. Significantly, causal and dilated convolutions are central concepts of TCN.

3.2.1. Causal convolutional network

A causal convolutions architecture (Figure 2) was adopted to address sequence modelling. The architecture uses a multi-layer fully-convolutional network, with the benefit of effectively learning and extracting the abstracted representation of input features through layer-wise processing.

3.2.2. Dilated convolutional network

Although multi-layer causal convolutions can expand the coverage of historical data, they are limited to looking back in history with a depth proportional to the network's linear timescale. Therefore, capturing a long history's information is challenging. To address this issue, dilated convolutions (Figure 2) was utilised to enable a large receptive field. Given filter function f with size k , the dilated convolution (F) for time series input x is defined as follows:

$$F(s) = (x *_{d} f)(s) = \sum_{i=0}^{k-1} f(i) \bullet x_{s-d \bullet i} \quad (2)$$

where s is the element of the sequence, d is the dilation parameter, and $s - d \bullet i$ describes the direction of the past.

3.2.3. Residual connections

Residual connections (Figure 3) were used throughout the network to speed convergence and enable training of much deeper models. A residual block contains a branch, which effectively allows layers to learn modifications to the identity mapping. Figure 3 shows a basic network layer (named Dilated Casual Layer) that includes 2 * (1D Dilated Convolution + Weight_Norm + Relu + Dropout) and a residual Convolution.

3.3. Safety assessment of rocket flight state

3.3.1. TOPSIS model

TOPSIS model, introduced by Hwang and Yoon [88], is applied to the safety assessment of rocket flight status. The basic idea of TOPSIS model is rather straightforward [90]. This methodology reduces multiple parameters into a single value to facilitate safety assessment of different working conditions [95]. It originates from the concept that a higher safety score should have the shortest distance from the positive-ideal working condition A^+ and the farthest from the negative-ideal working condition A^- [91]. The flight parameters matrix is defined as a $i \times j$ matrix (i times and j parameters) and the idea of TOPSIS model can be expressed in a series of steps [96]:

- 1) Converge the parameter X_j , get x'_{ij} ;

$$x'_{ij} = \frac{M_j}{M_j + |x_{ij} - M_j|} \quad (3)$$

Where M_j is the value of the best status for parameter M_j .

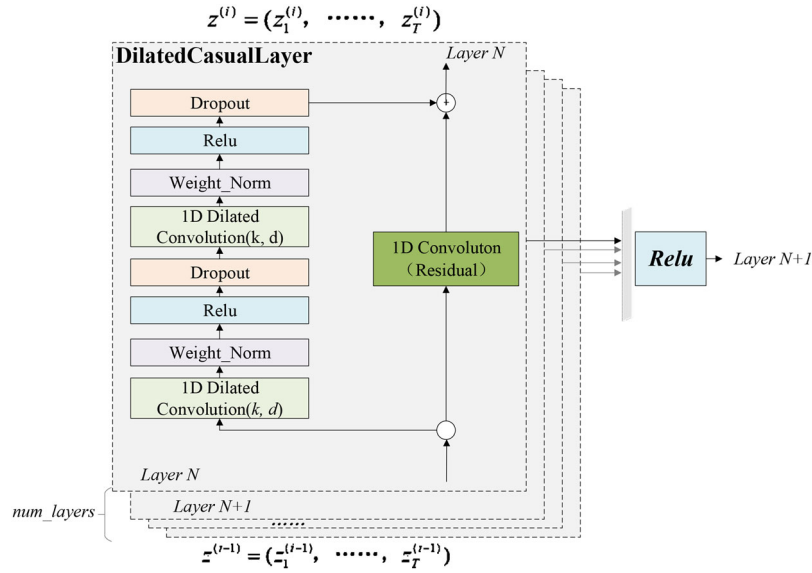


Figure 3. Architecture of a basic network layer of TCN (DilatedCasualLayer).

- 2) Normalise the condition matrix x'_{ij} , get a_{ij} ;

$$a_{ij} = x'_{ij} / \sqrt{\sum_{i=1}^m x'_{ij}{}^2}, a_{ij} \in (0, 1) \quad (4)$$

- 3) Develop a set of weights W ;

$$W = (\omega_1, \omega_2, \dots, \omega_n) \quad (5)$$

- 4) Determine positive-ideal condition A^+ and negative-ideal condition A^- ;

$$A^+ = (a_{i1}^+, a_{i2}^+, \dots, a_{in}^+), A^- = (a_{i1}^-, a_{i2}^-, \dots, a_{in}^-) \quad (6)$$

- 5) Develop the distance from present condition to both condition A^+ and condition A^- ;

$$D_i^+ = \sqrt{\sum_{j=1}^n \omega_j (a_{ij} - a_{ij}^+)^2}, D_i^- = \sqrt{\sum_{j=1}^n \omega_j (a_{ij} - a_{ij}^-)^2} \quad (7)$$

- 6) Determine the current safety score S_i :

$$S_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (8)$$

A bigger S_i indicates that the flight status is closer to the positive-ideal condition A^+ and farther from the negative-ideal condition A^- .

3.3.2. Entropy-weight method

Performing safety assessment requires objective weighting of parameters to ensure reliable results. The entropy-weight method is used for this purpose, as it is based on Shannon entropy and reflects the degree of disorder on a criterion. The higher the entropy, the lower the effectiveness of the corresponding information [97], and thus the smaller the parameter weight should be. This makes the entropy method an objective approach to determining weights in the model. Entropy is well-suited to measuring the relative intensities of contrast parameters and represents average intrinsic information [98], therefore, it is used to determine weights in step (3) for a collected condition matrix x'_{ij} . The attribute evaluation entropy is calculated as [99]:

$$E_j = -K \sum_{i=1}^m P_{ij} \ln(P_{ij}) \quad E_j \in [0, 1] \quad (9)$$

Where $P_{ij} = x_{ij} / \sum_{i=1}^n x_{ij}$, $K = 1 / \ln(n)$. Then, the attribute evaluation entropy weight is [100]:

$$W_j = \frac{d_j}{\sum_{j=1}^n d_j} \quad (10)$$

Where $d_j = 1 - E_j$.

Additionally, to incorporate domain expertise and practical engineering experience, the subjective adjustment parameter λ_i was determined through a structured expert consultation process. This process was designed to systematically capture expert judgement and mitigate individual bias. Five senior rocket safety engineers with over 10 years of field experience each were invited to participate in a Delphi method-based assessment.

- **3 senior research scientists** from the China Academy of Engineering Physics, each with over 15 years of experience in rocket dynamics and trajectory simulation.
- **2 chief engineers** from a leading aerospace manufacturer, each with over 12 years of hands-on experience in launch vehicle system design and flight test analysis.

The consultation followed a two-round, Delphi method approach to reach a consensus:

- **First Round (Independent Rating):** Each expert was provided with detailed definitions of all 12 flight parameters (Table 1). They were asked to independently rate the relative criticality of each parameter to the overall flight safety on a scale of 1 to 5, where 1 indicated ‘minimal importance’ and 5 indicated ‘critical importance.’ The ratings were collected anonymously.
- **Second Round (Consensus Building):** The anonymised ratings and justifications from the first round were compiled and shared with the entire panel. A facilitated meeting was then held where experts discussed the discrepancies, particularly for parameters with high rating variance. Following this discussion, experts were given the opportunity to revise their ratings anonymously based on the insights gained.

The final λ_i value for each parameter was calculated as the normalised mean of the ratings from the second round. This process ensured that the final subjective weights reflected a consolidated, consensus-based view. For instance, parameters with a direct and immediate impact on trajectory stability, such as X-position and X-acceleration, received higher consensus ratings, resulting in λ_i values of approximately 1.25. In contrast, parameters considered to have a more moderate influence, such as roll angle, received a λ_i value of approximately 0.95.

This structured approach ensured that both data-driven objectivity (from the entropy method) and validated practical engineering wisdom were incorporated into the final weights. The comprehensive weight W_j^0 for each parameter, combining both objective and subjective components, is therefore calculated as:

Table 1. Dataset description of parameters during flight.

Variable type	Variable name	Data type	Unit
Distance	X-position	Numeric	m
	Y-position	Numeric	m
	Z-position	Numeric	m
Acceleration	X-acceleration	Numeric	m/s ²
	Y-acceleration	Numeric	m/s ²
	Z-acceleration	Numeric	m/s ²
Speed	X-speed	Numeric	m/s
	Y- speed	Numeric	m/s
	Z- speed	Numeric	m/s
Flight attitude	Pitch angle	Numeric	°
	Raw angle	Numeric	°
	Roll angle	Numeric	°

$$W_j^0 = \frac{\lambda_j W_j}{\sum_{k=1}^n \lambda_k \cdot W_k} \quad (11)$$

where W_j is the entropy weight from Eq. (10), and λ_j is the subjective adjustment factor derived from the expert consultation.

3.3.3. Rocket safety status assessment with TOPSIS model

The flight envelope curve and ideal flight state curve can be obtained through flight trajectory simulation. When in time t , positive-ideal condition A^+ for parameter X_j is *Flight_envelope(t)* (red line in Figure 4), meanwhile positive-ideal condition A^+ for criteria X_j is *Theoretical_flight_state_curve(t)* (green line in Figure 4).

Figure 5 depicts the calculation process of the TOPSIS model. In step (3), the closer to the best status M , the healthier the status is. Here, $x'_{ij} = M / [M + |x_{ij} - M|]$. In step (4), a_{ij} increases from 0 to 1, indicating that the rocket safety status changes from negative-ideal condition A^- to positive-ideal condition A^+ . In step (5), the weights of parameters indicate their relative and overall importance in comparison to other parameters, which is a key component of the assessment result. It is typically defined by expert judgement and on-site investigation [101]. In this paper, entropy-weight method is used to help determining the weights. In step (6), negative-ideal condition A^- and positive-ideal condition A^+ are defined using pre-prepared simulation parameters, and are used for further safety assessment in step (7) and step (8).

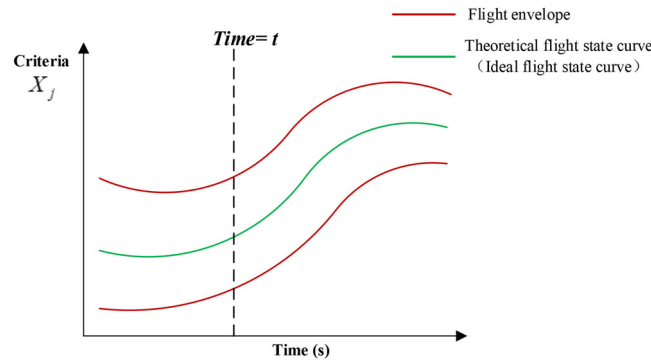


Figure 4. Negative-ideal condition A^- and positive-ideal condition A^+ for parameters X_j when in rocket flight experiments.

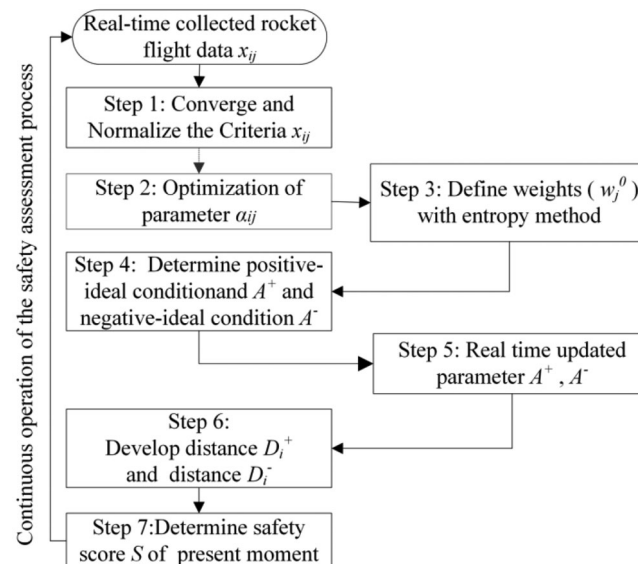


Figure 5. Schematic diagram of the TOPSIS model for rocket safety assessment.

It should be emphasised that the negative-ideal condition A^- and positive-ideal condition A^+ are updated in real-time with rocket flight time.

4. Case analysis

4.1. Data source and semi-physical simulation

The datasets used for model training and validation were generated through a high-fidelity semi-physical simulation platform developed in-house. This platform integrates mathematical models of rocket dynamics, propulsion, guidance, and control systems with physical hardware components, such as sensors and data acquisition systems, in a closed-loop testing environment. This approach, consistent with methodologies described in [47,48], creates a highly realistic testing scenario that bridges the gap between pure numerical simulation and full-scale flight tests.

The simulation was configured to replicate the flight profile of a representative solid-propellant launch vehicle. The mathematical models were calibrated and validated against historical flight data from three previous successful launch missions, ensuring that the simulated parameters (e.g. trajectory, dynamics, and attitude) exhibit consistent statistical characteristics and physical plausibility with actual flight data. Key performance indicators, such as the range and distribution of accelerations and attitude rates, were confirmed to be within the observed bounds of real flight experiments.

The scale of the generated dataset is substantial, comprising over 50 simulated flight tests. Each test covers the entire powered flight phase, resulting in multivariate time series of approximately 120 seconds in duration, sampled at 100 Hz. This corresponds to roughly 12,000 time steps per flight for each of the 12 parameters listed in Table 1. The dataset includes:

- **Nominal Flight Scenarios:** 30 simulations under ideal conditions and with minor, acceptable perturbations.
- **Fault Injection Scenarios:** 20 simulations incorporating the six specific fault types detailed later in Table 4, with varying fault magnitudes and initiation times to ensure a comprehensive representation of potential anomalies.

4.2. Data processing

The raw data acquired from semi-physical simulations underwent a comprehensive preprocessing pipeline to ensure quality and consistency for model training. The preprocessing steps included:

- **Missing value handling:** Less than 0.5% of data points were identified as missing, primarily due to sensor simulation dropouts. These were addressed using linear interpolation for continuous parameters and forward-fill for categorical parameters, ensuring temporal continuity in the time-series data.
- **Noise filtering and smoothing:** High-frequency noise was attenuated using a Savitzky-Golay filter with a window length of 15 and polynomial order of 3, which effectively preserved critical trajectory features while reducing stochastic fluctuations inherent in simulation data.
- **Normalisation:** All flight parameters were normalised to a $[0, 1]$ range using min-max scaling based on their theoretical operational boundaries to ensure equal contribution during model training. The normalisation parameters were derived exclusively from the training set to prevent data leakage into validation and test sets.
- **Temporal alignment and resampling:** Multi-source data streams were synchronised using precise timestamps and resampled to a consistent 100 Hz frequency through cubic spline interpolation, ensuring uniform temporal resolution across all parameters.
- **Sequence segmentation:** The continuous time-series data was segmented into overlapping sequences of 50 time steps with a sliding window of 10 steps, creating training samples that capture temporal dependencies while augmenting the effective dataset size for robust model training.

Table 2. Parameters of TCN network for rocket flight parameters prediction.

Parameter type	Parameter name	Parameter description	Parameter value
Structural parameters	input_size	Number of input data	4
	Layer number	Number of DilatedCasualLayer (Figure 3)	4
	channel_size	Channels generated by convolution between DilatedCasualLayers	10
	output_size	linear transformation output	1
Internal parameters	kernel_size	Size of Convolutional Kernel	2
	stride	Convolution stride	1
	dilation	Dilation parameter of Dilated Convolution	2
	out_channel	Channels generated by convolution of 1D convolution in each DilatedCasualLayer	10
Training parameters	Error indicator	Error between input and target output	MSE
	optimiser	Training optimiser	Adam
	Dropout	Probability of dropout	0.2
	Lr	Learning rate	0.005
	Epochs	Training times	1000

Parameters (Table 1) during flight can basically describe the safety status of a rocket. Given the historical data set of N parameters, the input features can be expressed as a 2D vector, $X \in R^{T \times N}$, where T and N represent the number of time steps and the flight parameters.

4.3. Parameter setting

4.3.1. Rocket flight parameters prediction

Parameters of TCN (Table 2) include structural, internal and training parameters. The selection of hyperparameters such as the number of DilatedCasualLayers and kernel size was guided by both architectural principles and empirical tuning.

The TCN architecture is designed to capture long-range dependencies in sequential data through dilated convolutions and residual connections. The receptive field of a TCN is determined by the number of layers L , kernel size k , and dilation base d , and is given by:

$$\text{Receptive Field} = 1 + (k - 1) \cdot \frac{d^L - 1}{d - 1} \quad (12)$$

In this implementation, a dilation base $d = 2$, kernel size $k = 2$, and layers $L = 4$ were adopted, resulting in a receptive field of 31 time steps. This configuration ensures that the model can capture sufficient temporal context from the rocket flight data, which is critical for accurate prediction. The choice of 4 DilatedCasualLayers and a kernel size of 2 is consistent with recommendations from Bai et al. [43], showing that such configurations strike a balance between model capacity and computational efficiency. Similar settings have been successfully applied in time-series prediction tasks in industrial and mechanical systems [44–46].

To further validate those hyperparameter choices, a grid search over a range of values for the number of layers (2 to 8) and kernel sizes (2 to 9) was conducted. The final values for the number of DilatedCasualLayers (4) and the kernel size (2), as listed in Table 2, were selected following a comprehensive hyperparameter search. This search aimed to identify the configuration that optimally balanced prediction accuracy, measured by Root Mean Square Error (RMSE), with computational efficiency, which is critical for real-time assessment. The results of this search, detailed in Section 4.5.3 and visualised in Figure 11, demonstrate that this specific combination provides an excellent trade-off, achieving low RMSE while maintaining the fast computation time necessary for the digital twin system.

A TCN network with 4 Dilated Casual Layers (Figure 6) was constructed to produce rocket flight parameters prediction. Mean square error (MSE) was selected as the loss function for the training process. Adam optimiser was used for gradient-descent optimisation of the stochastic objective function. Concurrently, a learning rate (lr) of 0.005 and 1000 training epochs were utilised.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2 \quad (13)$$

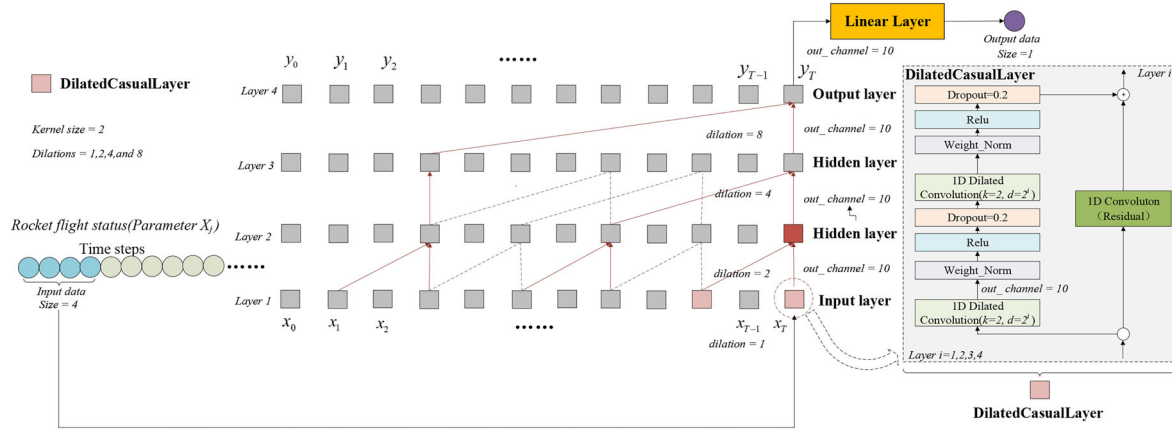


Figure 6. Architecture of TCN network designed in this paper.

Table 3. Weight values of rocket flight parameters.

Parameter	X-position	Y-position	Z-position	X-acceleration	Y-acceleration	Z-acceleration
Weight value	0.097	0.080	0.083	0.092	0.083	0.090
Parameter	X-velocity	Y-velocity	Z-velocity	Pitch angle	Raw angle	Roll angle
Weight value	0.079	0.076	0.076	0.086	0.078	0.079

4.3.2. Safety assessment

The entropy-weight method was used to determine the weights of rocket flight parameters. Effective weights were derived from both collected data and the previous engineering experience, as in Table 3.

When evaluating safety status during rocket flight, the positive-ideal condition A^+ (green line in Figure 7) is ideal parameter values obtained through ideal flight simulation. Flight envelope curves (red line in Figure 7) of different parameters are used to conduct on-site safety status analysis. The flight envelope curves can be considered as the negative-ideal condition A^- . Accordingly, the positive-ideal condition A^+ and negative-ideal condition A^- of different parameters can be shown in (Figure 7).

4.4. Model performance

4.4.1. Rocket flight parameters prediction

In this section, the effectiveness of the proposed model was validate. The Rocket flight parameters (Table 1) were predicted using the described TCN network. The Mean Square Error (MSE) was used as the model error evaluation indicator to assess performance. The lower the MSE value, the higher the prediction accuracy.

Figure 8 shows the MSE of three parameters (X-position/Y-position/Z-position) during the training process. It can be seen that the MSE value gradually converges and decreases to near 0, verifying the feasibility of the described TCN network in capturing temporal variations. Figure 9 shows the prediction results of rocket flight parameters using the TCN network. All of the results indicate that the TCN network performed well in rocket flight parameters prediction, and the prediction results are satisfactorily accurate.

4.4.2. Safety assessment

Several data sets of rocket flight parameters with different safety status were obtained through a semi-physical simulation, shown as Table 4. They were used to test the performance of the TOPSIS model described in this paper.

Figure 10 presents assessment scores computed from data sets in Table 4. As the first-stage engine failure occurs, the rocket safety assessment score decreases progressively (Fault Flight State Type 4, Fault Flight State Type 5, Fault Flight State Type 6), occurring earlier than the second-stage engine malfunction. As the rocket flight progresses, the safety assessment score gradually decreases below the safety threshold of 60 points. The assessment results align with the actual safety status of the rocket,

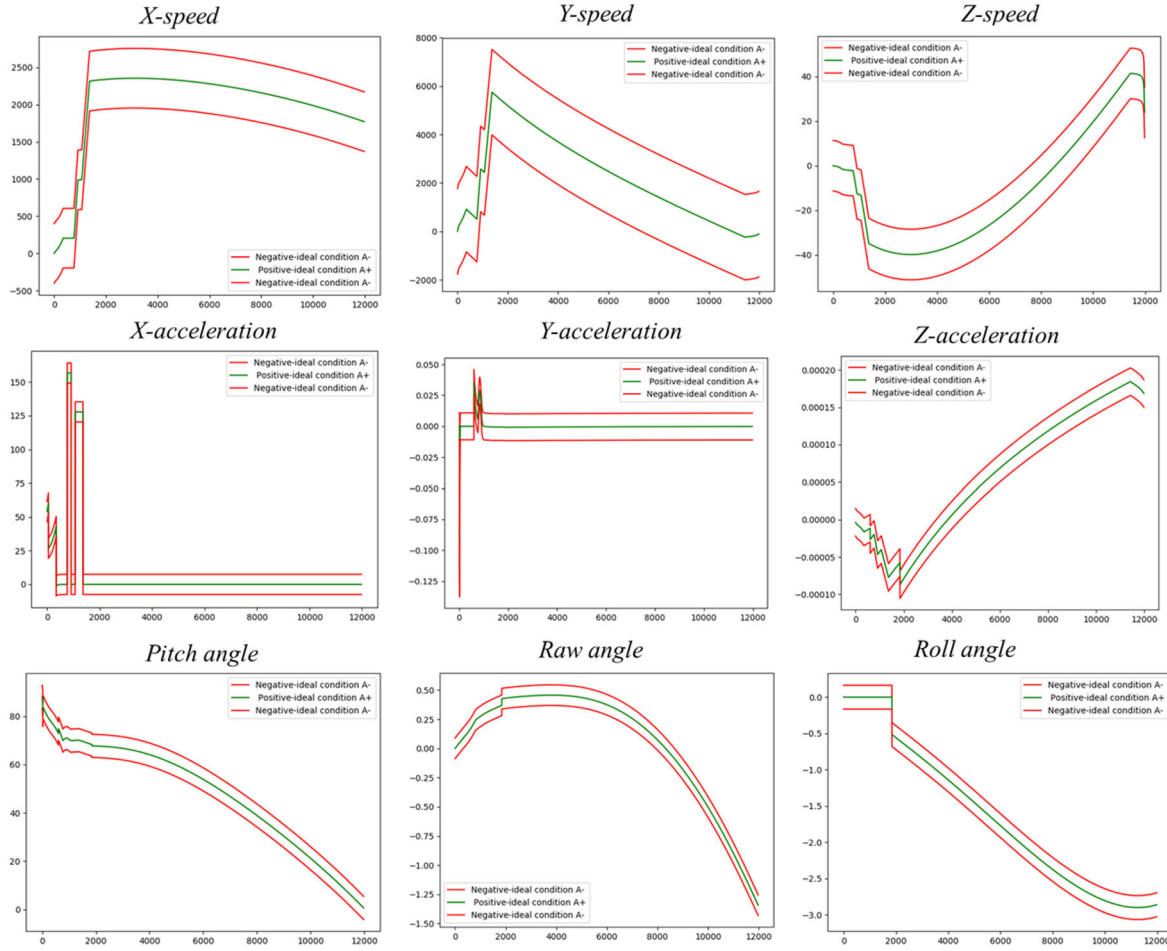


Figure 7. Positive-ideal condition and negative-ideal condition of parameter position and speed (red line: negative-ideal condition, green line: positive-ideal condition).

and the single overall value derived from the multiple criteria helps engineers quickly determine the safety status and make decisions promptly.

4.5. Discussion of results

4.5.1. Comparison with other methods

To ensure a fair and reproducible comparison of model performance, all experiments were conducted under identical hardware and software conditions. The unified experimental environment consisted of a CPU @ 2.20 GHz, 64 GB RAM, and a GPU with 10 GB VRAM. The software environment utilised Python 3.8, PyTorch 1.9.0.

All models (TCN, RNN, GRU, and LSTM) were trained and evaluated using consistent settings to eliminate confounding factors. Specifically:

- **Dataset split:** The same semi-physical simulation dataset described in [Section 4.1](#) was used, with an identical 70:15:15 split for training, validation, and testing, respectively.
- **Training parameters:** All models were trained for 1000 epochs using the Adam optimiser with a fixed learning rate of 0.005 and Mean Squared Error (MSE) as the loss function.
- **Batch processing:** A consistent batch size of 64 was applied across all models during both training and inference phases.
- **Input/output configuration:** All models received identical input sequences (historical flight parameters) and were configured to predict the same target parameters under the same sequence length settings.

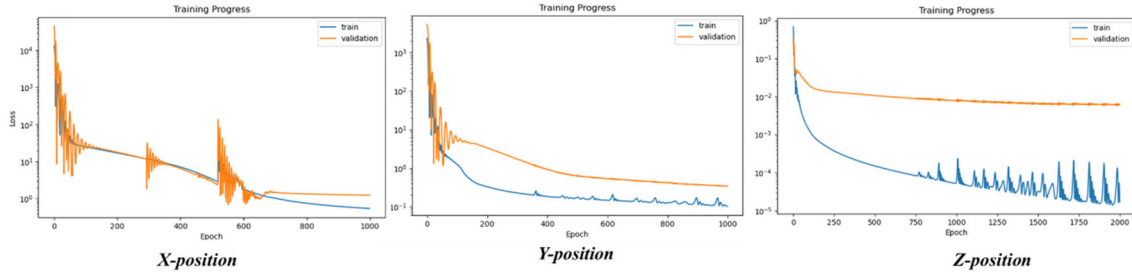


Figure 8. MSE of parameters (X-position/Y-position/Z-position) during the TCN network training process.

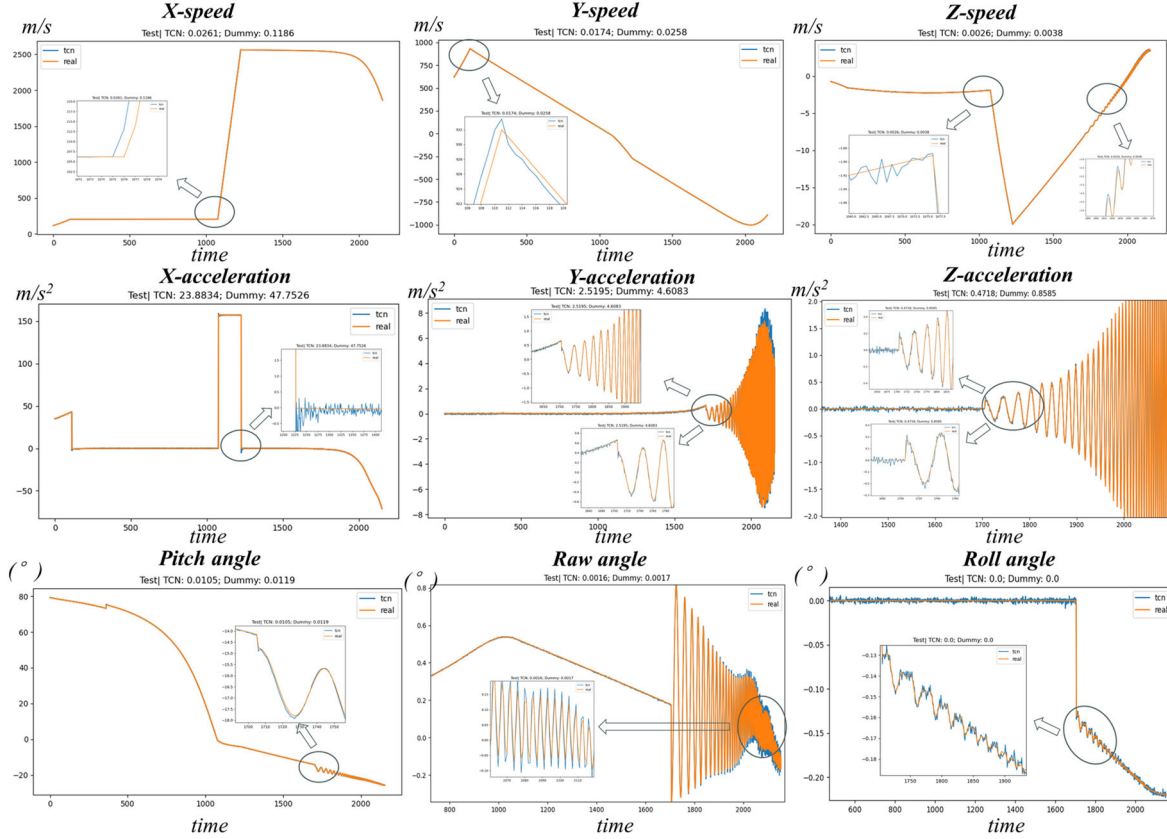


Figure 9. Rocket flight parameters prediction results by TCN network.

Table 4. Data sets of different safety states of rocket flight status.

Data set name	Data set introduction
Ideal Flight State Process	The rocket flies according to ideal conditions throughout the entire process.
Fault Flight State Type 1	The second-stage engine of rocket did not separate properly.
Fault Flight State Type 2	Abnormal increase in lateral thrust of the second-stage engine.
Fault Flight State Type 3	Premature Termination in thrust of the second-stage engine.
Fault Flight State Type 4	The first-stage engine of rocket did not separate properly.
Fault Flight State Type 5	Abnormal increase in lateral thrust of the first-stage engine.
Fault Flight State Type 6	Premature Termination in thrust of the first-stage engine.

The architectural complexities of the comparative models were designed to be approximately equivalent in parameter count to ensure a balanced comparison. The RNN, GRU, and LSTM models each contained two hidden layers with 50 units per layer, representing a complexity comparable to our 4-layer TCN architecture.

Table 5 shows the comparison results of the different models. TCN outperforms the other three networks (RNN, GRU, and LSTM) in terms of prediction accuracy (RMSE) and computational speed. This result suggests that the TCN network achieves better performance in safety prediction for rocket flight digital twin systems.

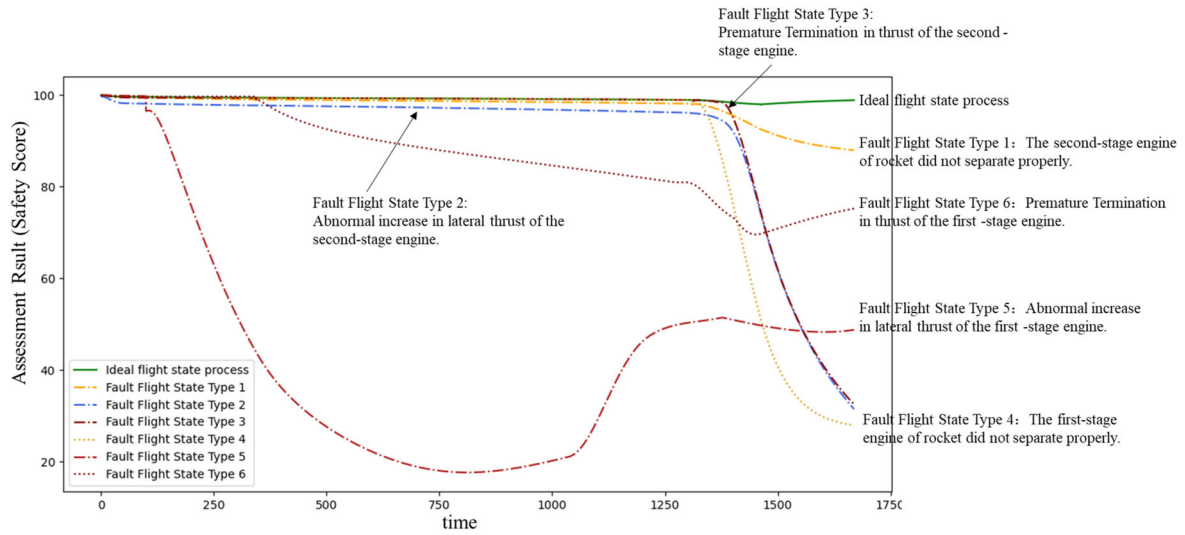


Figure 10. Assessment results of rocket flight safety status based on TOPSIS model.

Table 5. Model performance of TCN and comparisons with other models.

	RNN	GRU	LSTM	TCN
RMSE	0.18922	0.17820	0.45066	0.16651
Computation time (ms)	3.989	12.966	18.955	2.992

Table 6. Model performance of TCN and comparisons with other models.

	Hybrid CNN-LSTM [32]	CNN-GRU [35]	Att-BiLSTM [36]	TCN-TOPSIS
RMSE	0.18234	0.17589	0.16995	0.16651
Computation time (ms)	15.221	14.883	22.547	2.992

4.5.2. Comparison with state-of-the-art methods

To further validate the performance of the proposed TCN-TOPSIS model, its prediction accuracy and computational efficiency were benchmarked against several state-of-the-art sequence prediction models identified in the literature. These include advanced architectures that have recently been applied to similar time-series forecasting tasks in aerospace and safety-critical systems. Specifically:

- A Hybrid CNN-LSTM model [32,33], which leverages CNNs for spatial feature extraction and LSTMs for temporal dependencies.
- A CNN-GRU model [35], which combines convolutional layers with Gated Recurrent Units.
- An Attention-based Bidirectional LSTM (Att-BiLSTM) [36], which utilises attention mechanisms to weigh the importance of different time steps and a bidirectional structure to capture past and future context.

All comparative models were implemented and trained under the same unified experimental environment and dataset described in Section 4.4.1 to ensure a fair comparison. The results, summarised in Table 6, demonstrate that the proposed TCN-TOPSIS model achieves a superior balance between high prediction accuracy (lowest RMSE) and low computational latency. While the Att-BiLSTM model achieved competitive accuracy, its computational time was significantly higher due to the complexity of the attention mechanism and bidirectional processing, making it less suitable for real-time rocket flight assessment. The hybrid CNN-LSTM and CNN-GRU models also exhibited higher RMSE compared to our TCN-based approach.

The superior performance of TCN-TOPSIS can be attributed to the parallelizable architecture of the TCN, which provides a larger effective receptive field with fewer parameters and more efficient computation compared to recurrent and hybrid models. This empirical evidence, consistent with findings from Bai et al. [43], positions TCN-TOPSIS as a leading candidate for real-time safety prediction in digital twin systems where both accuracy and speed are paramount.

4.5.3. Parameter influence on the model performance

Two critical parameters of TCN, layers and kernel size, strongly influence predictor performance. The number of layers defines the network architecture, while the kernel size describes the receptive field for convolution operations. These two parameters (described in Section 3) greatly affect the prediction accuracy and response time of the TCN network.

Figure 11 shows the performance of TCN under different combinations of the two parameters. The layer size was set from 2 to 8, and the kernel size was set from 2 to 9. The two parameters in the network affected both model accuracy and computation time, and a balanced combination can achieve

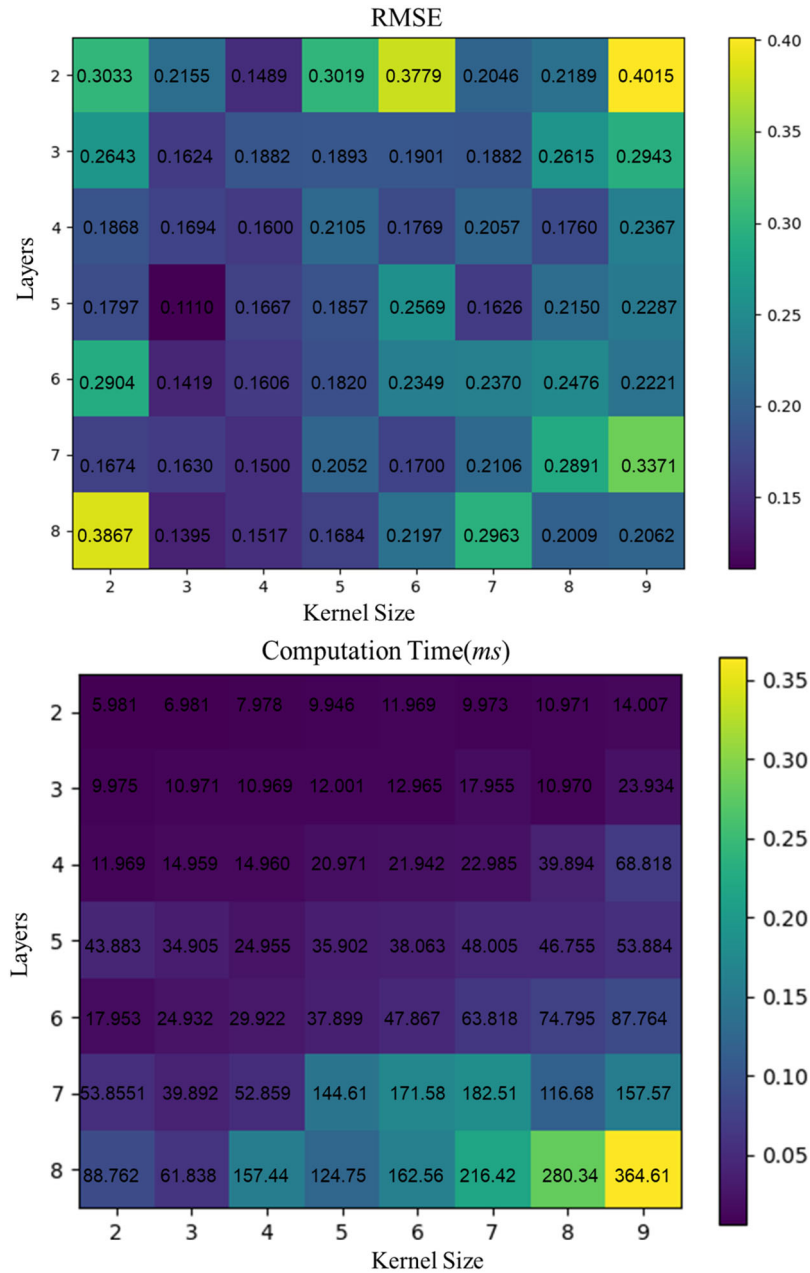


Figure 11. RMSE and computation time with different combinations of kernel size and layers.

good results. As illustrated in Figure 11, the combination of a kernel size of 2 and 4 layers was ultimately selected as the optimal configuration for our TCN model, as it effectively balanced the competing demands of high predictive accuracy (low RMSE) and minimal computational latency, fulfilling the core requirement for real-time rocket safety assessment.

4.5.4. Complexity and computational time analysis

A thorough analysis of computational complexity and execution time was conducted to evaluate the efficiency of the proposed TCN-TOPSIS framework, which is crucial for real-time rocket safety assessment applications.

The computational complexity of the TCN model is primarily determined by its convolutional operations. For an input sequence of length T , filter size k , number of layers L , and number of filters C , the theoretical time complexity of TCN is approximately $O(L \cdot k \cdot C \cdot T)$, which grows linearly with sequence length. This represents a significant advantage over recurrent architectures like LSTMs and GRUs, which have complexity $O(L \cdot C^2 \cdot T)$ due to their sequential processing nature and dense matrix operations. The TOPSIS evaluation component has complexity $O(m \cdot n)$ for m alternatives and n criteria, making it highly efficient for real-time safety scoring.

To provide granular insights into system performance, the execution time of each major component during inference was measured (averaged over 1000 runs):

- TCN Feature Extraction: 1.82 ± 0.23 ms;
- Parameter Prediction: 0.58 ± 0.11 ms;
- TOPSIS Safety Assessment: 0.61 ± 0.09 ms;
- Total Inference Time: 3.01 ± 0.32 ms.

This breakdown confirms that the complete TCN-TOPSIS pipeline operates well within the stringent real-time requirements of rocket flight safety systems, with the TCN prediction dominating the computational load while still maintaining high efficiency.

5. Conclusions

This study developed a TCN-TOPSIS model for rocket safety state prediction. The proposed model combines a TCN network with a TOPSIS evaluation model. The TCN-TOPSIS model exhibits accurate prediction results and fast calculation speed, making it highly suitable for incorporation into rocket flight digital twin systems.

1. In this article, TCN is utilised to predict rocket flight status, combining the parallel processing capability of convolutional neural networks with the long-term dependency modelling capability of recurrent neural networks (RNNs). The core structures of TCN include one-dimensional convolution, causal convolution, dilated convolution, and residual connections, which enable TCN to effectively capture long-term dependencies while maintaining computational efficiency when processing sequential data. In the prediction of rocket flight state parameters, TCN outperforms RNN, GRU, and LSTM in terms of prediction accuracy (RMSE) and computational speed. Two important parameters of TCN, layers and kernel size, strongly affect prediction performance. After detailed comparative analysis, the optimal combination of parameters is 4 for kernel size and 4 for layers.
2. The TOPSIS model is utilised for real-time assessment of rocket safety status. Objective criterion weights are obtained using the entropy weight method. The positive-ideal (A^+) and negative-ideal conditions (A^-) are employed for real-time safety assessment. TOPSIS model reduces multiple criteria into a single value, facilitating safety assessment of different working conditions. Experimental results demonstrate that the TOPSIS model exhibits fast calculation speed and accurately assesses the safety status of flying rockets. It can be concluded that the proposed TCN-TOPSIS model is a potentially effective and accurate method for rocket safety state prediction in digital twin systems.

Despite the promising results, this study has certain limitations that suggest directions for future research. Firstly, the current model was trained and validated primarily on data from semi-physical simulations and nominal fault scenarios. Its performance under extreme, out-of-design-range flight conditions (e.g. multiple simultaneous subsystem failures or highly nonlinear aerodynamic effects) requires further investigation. Secondly, the feature selection for the model was based on physical principles and domain expertise; a formal statistical analysis of inter-parameter correlations and causal relationships within the dataset was not performed. Conducting such analyses in future work could provide deeper insights into feature dependencies and potentially lead to further model optimisation. Finally, the TCN and TOPSIS model parameters were optimised for a specific rocket configuration; adaptation to significantly different vehicle designs may necessitate retuning and validation. Future work will focus on: (1) validating the model with real flight data across a wider envelope of operational and failure conditions; (2) developing an adaptive framework that can automatically adjust model parameters or structures in response to changing vehicle characteristics or mission profiles; and (3) exploring hybrid modelling approaches that integrate physical knowledge with data-driven methods to enhance extrapolation capability and interpretability, particularly for rare or unforeseen fault modes.

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Authors' contributions

CRedit: **Fugang Li**: Formal analysis, Methodology, Writing – original draft; **Bo Li**: Data curation, Formal analysis, Resources; **Lei Zhao**: Methodology, Software; **Hao Li**: Methodology, Writing – review & editing; **Yuehui He**: Formal analysis, Software; **Junsheng Zheng**: Data curation, Investigation, Software; **Yonglin Bao**: Conceptualisation, Resources, Writing – review & editing; **Zhiwei Ding**: Formal analysis, Project administration, Writing – review & editing.

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